How-To Library

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Implementing Event Tables to Control Extraction Processing by PowerCenter CDC Sessions

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Abstract

You can use a PowerExchange event table to stop the extraction of change data at a specific point in the real-time change data stream or the PowerExchange Logger or PowerExchange Condense files. This feature is useful if you need to stop CDC sessions before or after running routine batch jobs or before performing maintenance.

Supported Versions

- PowerExchange 10.0 or later

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Overview of Event Tables

You can use a PowerExchange event table to extract change data up to a specific point in the change data stream and then stop the associated PowerCenter CDC real-time session. An event table provides a way of creating a marker in the change stream relative to the execution of other time-sensitive processes. For example, you can use an event table to stop the CDC session after it processes all of the changes up to the end of the business day.

You can implement an event table for any CDC data source type. For nonrelational data sources, the event table is not a relational table. For example, it can be an Adabas data set, IMS data segment, IDMS record, or VSAM file. For simplicity, this article uses the term *event table* to refer to this feature for all CDC data source types. Also, the term *change stream* refers either to the changes from real-time change capture or from PowerExchange Logger log files or PowerExchange Condense files.

You must create the event table on the source system and configure the CDC connection to use it. You must also define a process that creates a marker in the event table to identify the point in the change stream where you want extraction processing and the CDC session to stop.

When the PowerExchange Client for PowerCenter (PWXPC) reader detects an event marker in the change stream, it adds an end-of-file (EOF) record. PWXPC passes the EOF record to the PowerCenter Integration Service. The PWXPC CDC Reader then shuts down. After the CDC session finishes processing the change data for the current UOW, the session also stops.

Note: When an event table is in use by an extraction session that has a real-time CDC source, such as a DB2 for i5/OS source or possibly a DB2 or SQL Server source on Linux, UNIX, or Windows, PowerExchange does not continue to capture changes from the source after the session stops. For other CDC sources, such as those on z/OS, PowerExchange continues to capture changes from the source database after the session stops. In this case, the event table controls only the extraction of the captured change data by the related CDC session.

By using an event table, you can stop change data extraction and a CDC session based on a specific point in the change stream, as needed to meet your business processing requirements. If you stop the workflow based on a clock time instead, extraction processing might stop too early in the change stream to get all of the captured changes needed for end-of-day (EOD) processing or other time-sensitive batch processing.

The following sample scenarios describe some uses of event tables:

- You need to implement source schema changes. You use an event table to stop the CDC extraction session after the session has consumed all of the daily changes and before the maintenance job that implements the schema changes runs.

- You need to update a data warehouse with all insurance policy changes from the operational system once a day so that the data warehouse reflects the EOD position of the policies. Before the CDC session processing is complete, the online system must be available for the next day's business. In this case, you use an event table to stop the CDC extraction session at the EOD event. The online system continues to generate changes, and PowerExchange continues to capture those changes. The changes captured after the EOD event will be extracted and sent to the data warehouse the next day.
- You want to extract all changes up to the end of nightly batch processing. To do so, you add a process that updates the event table as the last step of the nightly batch job. When the CDC session detects the event marker in the change stream, the session shuts down.

Requirements for Event Table Use

To use event tables, ensure that the following requirements are met:

- You can use only one event table for a CDC session.
- You must extract change data in real-time or continuous extraction mode.
- The event table and all of the source tables in the CDC session must be of the same data source type and reside on the same machine. For example:
  - An Oracle event table must reside in the same Oracle instance as the Oracle source tables.
  - A DB2 for z/OS event table must reside in the same DB2 subsystem or DB2 data sharing group as the DB2 source tables.
  - A VSAM event table (file) must reside on the same LPAR as the VSAM source data sets.
- You must use the same PWXPC CDC Real Time application connection to access the event table as you use to access the source tables that are defined in the CDC session.

Implementing an Event Table

Complete this procedure to create an event table and use it for a new or existing PowerCenter CDC session.

1. Create the event table.

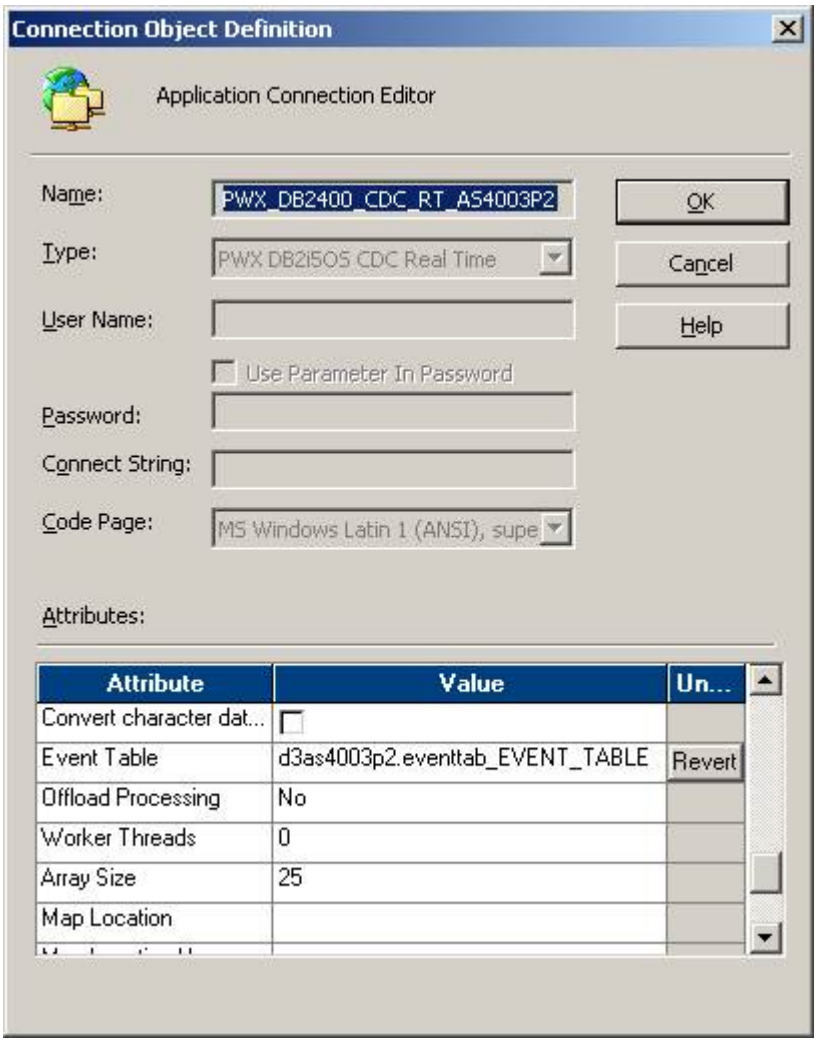
   The event table must be of the same data source type as the CDC data source and reside on the same machine.

   Tip: A simple event table could consist of two columns: 1) a single TIMESTAMP column that stores the current timestamp when an event marker is generated, and 2) a character or text column that stores the reason for the event table record. By including the reason, you can use the event table for auditing purposes and to get an historical view of why extractions were stopped.
2. In the PowerExchange Navigator, create a capture registration for the event table.

   The PowerExchange Navigator automatically generates a corresponding extraction map. The extraction map name has the format *schema.map_tablename*.
3. In PowerCenter Workflow Manager, create the CDC session and a PWX CDC Real Time application connection.

   Alternatively, you can define an event table for an existing CDC session and connection.
4. Open the PWX CDC Real Time application connection that you defined for the CDC session.
5. In the **Connection Object Definition** dialog box, specify the name of the extraction map in the **Event Table** attribute.

The following image shows the **Event Table** attribute:



6. Create a process to update the event table with a marker that identifies the point in the change stream where extraction processing should stop.
7. If you are implementing the event table for an existing CDC session rather than a new session, stop the CDC session and restart it.

Author

Virginia Pfeifle